IM3080 Design and Innovation Project (AY20xx/xx Semester x)
Individual Report

Name: Ng Yan Qing

Group No: 5

Project Title: Jack and the Beanstalk

Contributions to the Project (1 page)

Finance: In charge of all financial matters including claims.

Overall design: Created majority of the initial design sketches, including the addition of colour after we decided on our mood board. I also oversaw the designing and building of the main outline of our project, ensuring that everything was placed where it was with a purpose, including the building of the beanstalk. Also helped out in other parts such as drilling holes in the supporting tube for the electronics

Well (coin-filtering): To make the interactive starlight more "interactive", I had thought that it would be more interesting if we were to add in a coin filtering system into the well such that it seemed more like a story-telling design, such that the lightshow would only be triggered upon payment to the storyteller (bunny on the well). The filter is made entirely of cardboard, with no electronic parts required and took me multiple trial and errors before I was able to find the best solutions. I learnt to make use of whatever items I had on hand to come up with the solutions.

Songs: The first version of the songs was a mashup of 2 songs. We had later decided to change to a single song for the entire lightshow. However, after further discussions, we decided to go back to the initial idea which had more of a storytelling element to it. It took me a long time to find different pieces to remix and mash together, especially since none of the group members have musical background and the 2 songs I had found for the first part had very different tempos, making them clash. I had to spend quite some time playing around with the software to figure out how to better mash the music together. From this, I have learnt that it is okay to have the ideas go around in a circle and end back in the starting point as the new ideas may not always turn out as expected.

Wire Mesh/Flower LED: Ideated the use of wire mesh to hold the flowers in position while also providing space for future maintenance and also provide volume to the clouds. The clouds also provide a perfect cover for the electronics at the top, while the wire mesh holds them in place. It was difficult to ensure that the wire mesh was folded in ways such that it remained 3-dimensional, functional and easily attachable. I also made sure that the petals were attached to the LEDs properly and ensured that each one was of quality while remaining unique.

Servo Motor: Assisted in the building and testing of the servo mechanism. As the parts that we had purchased were not of quality and could only be purchased in bulk, I had to request for some better quality Lego parts in small quantities from external sources such that we were able to better secure the gearbox.

Reflection on Learning Outcome Attainment

Reflect on your experience during your project and the achievements you have relating to <u>at least two</u> of the points below:

(a) Engineering knowledge
(b) Problem Analysis
(c) Investigation
(d) Design/development of Solutions
(e) Modern Tool Usage
(f) The Engineer and Society
(g) Environment and Sustainability
(h) Ethics
(i) Individual and Team Work
(j) Communication
(k) Project Management and Finance
(l) Lifelong Learning

Point 1: <u>Project Management and Finance</u>
Being placed in charge of all the financial matters of the group ensured that I was up to date with the progress and decisions made by the different team. Tracking the purchases and submitting claims were definitely not as smooth sailing as I expected due to the many requirements before we could even purchase an item and submit a claim. Having to keep everything within budget was difficult especially when there were initially plans on changing to raspberry pi instead, where each piece would have used up over half the budget.

Additionally, the management of the project was much harder than expected despite only being pseudo-in-charge of the design team as it was the first time I had worked with such a big team, with everyone working on different parts of the project. Yu Sheng has definitely done a good job as our leader, coordinating the different groups together. This has helped made it easier for me to keep up with the progress of the different teams (design and electronics), which allowed for a smoother communication of ideas between everyone.

Point 2: <u>Design/Development of solutions</u>
Many problems arouse throughout the whole project which required us to constantly analyse the issues faced before we could find alternatives. Having to analyse the main causes of problems, to finding out possible solutions and testing them out has proven to not be easy.

One such example would be the sorting of the coins. Having gained inspiration from YouTube videos of cardboard vending machines, I had thought that it would be rather simple to replicate it. However, I found out later that the sizes of Singapore's coins were too similar to be accurately separated when I cut out the hole to size by hand. Additionally, due to the limited space within the well, it was difficult to ensure that the coins will slide properly and not stop halfway through. As such, I had to create multiple prototypes and test them out before I could find the best solution (which was to cut the hole in a diamond shape). Initially, I had thought that the coin sorting would be very easy and fast to make, however, I was immediately humbled from the hands-on attempt when I realised not everything would work out as pictured.

Other issues also occurred throughout such as the servo motor being too weak to lift the clay model jack and constantly being detached from the gearbox, as well as the songs initially chosen were not to the liking of the team. However, I had managed to help solve these issues through various methods such as obtaining items to better secure the servo motor, as well as finding and editing the music such that it is of standard and accepted by the team.